PHASE 1 • BRAINSTORMING & IDEATION

✓ VERIFIED SPECIFICATION

Define Problem Statements

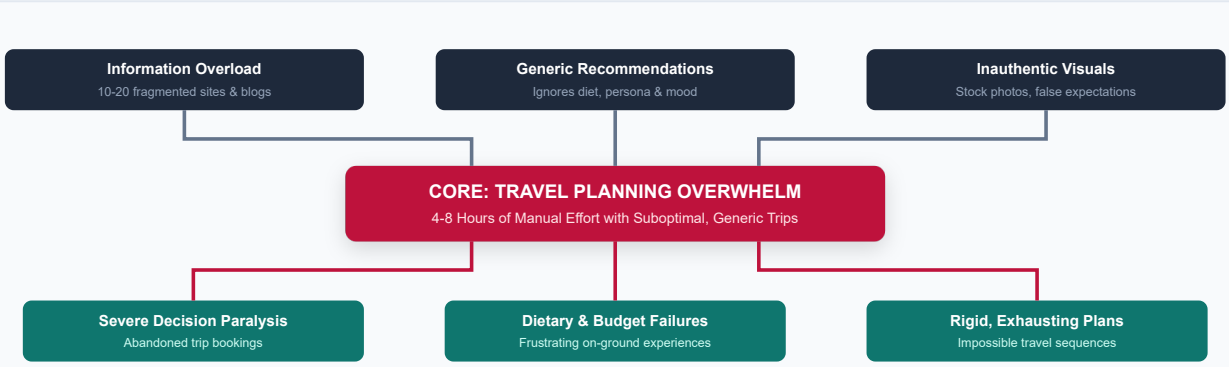
AI-Powered Travel Planner • Smart Situational Itinerary Engine

Live URL: travel-planner-iota-plum.vercel.app **Repo:** github.com/itskian2007/travel-planner **Tech:** React + FastAPI + PostgreSQL **Team:** itskian2007

1. Problem Discovery & Root Cause Analysis

Extensive user interviews with frequent leisure and business travellers revealed an industry-wide paradox: despite the abundance of online travel information, creating a cohesive, personalised trip itinerary remains an exhausting, friction-heavy endeavor. The core issue stems from **fragmented data silos**, **lack of situational intelligence**, and **generic recommendation algorithms** that ignore individual dietary, mood, and budget constraints.

2. Problem Tree Diagram (Cause • Core • Effect)



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3. Formal Problem Statements

PS-01: Cognitive Fragmentation — CRITICAL

Statement: Modern leisure travellers spend an average of 6.2 hours across 14 independent platforms to formulate a basic itinerary, leading to high cognitive fatigue and a 42% cart/plan abandonment rate.
Target Persona: Solo professionals, time-constrained couples.

PS-02: Demographic Blindness — HIGH

Statement: Existing travel engines recommend generic tourist attractions without factoring in dietary restrictions (Halal, Vegan), accessibility constraints (wheelchair access), or travel persona dynamics (solo vs. family).
Target Persona: Vegan/Halal travellers, families with elderly.

PS-03: Temporal Scalability Deficit — HIGH

Statement: Most travel guides cover only 2-3 standard days of sight-seeing. When travellers book 7 to 10 day vacations, they are forced to manually invent daily routes without verified geographic sequencing.
Target Persona: Extended vacationers, international backpackers.

PS-04: Visual Inauthenticity — MEDIUM

Statement: Commercial travel sites display heavily filtered stock photos or repetitive duplicate images, leading to on-ground expectation mismatches and tourist dissatisfaction.
Target Persona: Photography enthusiasts, cultural travellers.

4. How Might We (HMW) Design Framework

HMW CODE	REFRAMED DESIGN CHALLENGE	ENGINEERED SOLUTION FEATURE
HMW-01	How might we compress 6 hours of travel research into a 30-second automated generation flow?	Deterministic Procedural Engine
HMW-02	How might we incorporate dietary, budget, and accessibility needs into every meal and stop?	5D Situational Filter Matrix
HMW-03	How might we ensure 100% authentic visual representation for every attraction?	Wikipedia API 304 Unique Media Pipeline
HMW-04	How might we support realistic 10-day itineraries without boring repetition?	Curated Day 1-3 + Procedural Day 4-10